

A printer

Goals

We now have a scanner and the bones of a parser that makes a syntax tree

To make it easier to debug, let's convert that syntax tree into lines that you can print

```
1  (* (- 123) (group 45.67))
```

The printer

```
1  lox/AstPrinter.java
2  create new file
```

```
1  package com.craftinginterpreters.lox;
2
3  class AstPrinter implements Expr.Visitor<String> {
4      String print(Expr expr) {
5          return expr.accept(this);
6      }
7  }
```

Simply using the visitor pattern to walk the tree and return a string

The printer

```
1  lox/AstPrinter.java
2  add after print()
```

```
1  @Override
2  public String visitBinaryExpr(Expr.Binary expr) {
3      return parenthesize(expr.operator.lexeme, expr.left, expr.right);
4  }
5  @Override
6  public String visitGroupingExpr(Expr.Grouping expr) {
7      return parenthesize("group", expr.expression);
8  }
9  @Override
10 public String visitLiteralExpr(Expr.Literal expr) {
11     if (expr.value == null) return "nil";
12     return expr.value.toString();
13 }
14 @Override
15 public String visitUnaryExpr(Expr.Unary expr) {
```

The printer

```
1  lox/AstPrinter.java
2  add after visitUnaryExpr()
```

```
1  private String parenthesize(String name, Expr ... exprs) {
2      StringBuilder builder = new StringBuilder();
3
4      builder.append("(").append(name);
5      for (Expr expr : exprs) {
6          builder.append(" ");
7          builder.append(expr.accept(this));
8      }
9      builder.append(")");
10
11     return builder.toString();
12 }
```

The printer

Some Test data since we don't have a parser yet

```
1  lox/AstPrinter.java
2  add after parenthesize()
```

```
1  public static void main(String[] args) {
2      Expr expression = new Expr.Binary(
3          new Expr.Unary(
4              new Token(TokenType.MINUS, "-", null, 1),
5              new Expr.Literal(123)),
6          new Token(TokenType.STAR, "*", null, 1),
7          new Expr.Grouping(
8              new Expr.Literal(45.67)));
9
10     System.out.println(new AstPrinter().print(expression));
11 }
```

The printer

```
1 (* (- 123) (group 45.67))
```